RHCSA SA1

SYSTEM ADMIN

CHALLENGE

Submitted by

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Ipsr Cloud&DevOps(11:30)

Phase 1

Step 1: Downloading and Adding the operating system

```
Astra Linux 9.7
Hint: Num Lock on

Astra login: root
Password:
Last login: Thu Aug 20 11:24:40 on tty1
[root@Astra ~]#
```

Step 2: Configure the IP Address Manually

-Network configuration of the main os.

-system is using ipv4

```
C:\Users\hrish>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::fdf7:5916:bc78:b7%3
   IPv4 Address. . . . . . . . . . . : 192.168.59.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3cbe:3164:380b:4bcc%15
   IPv4 Address. . . . . . . . . . . : 192.168.157.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:8fde:db1c:4898:3594:8278:afd2
   Temporary IPv6 Address. . . . . . : 2401:4900:8fde:db1c:11d6:402c:7501:1738
   Link-local IPv6 Address . . . . . : fe80::c0c3:7d91:2f41:3b64%18
   IPv4 Address. . . . . . . . . . . : 192.168.1.20
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%18
                                       192.168.1.1

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\hrish>
```

```
[root@Astra ~]# ip route
default via 192.168.1.1 dev ens160 proto dhcp src 192.168.1.68 metric 100
192.168.1.0/24 dev ens160 proto kernel scope link src 192.168.1.68 metric 100
[root@Astra ~]#
```

-ip route of the Astra Linux in the vmware.

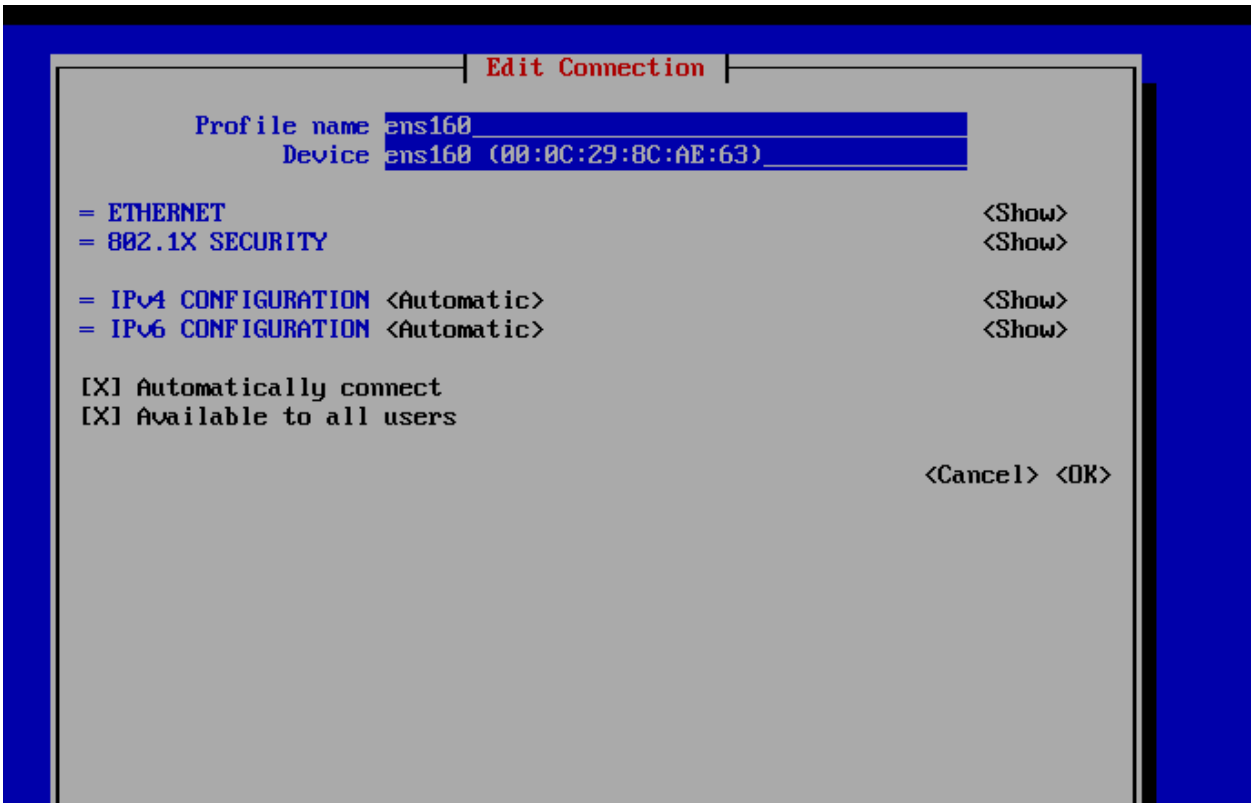
-checked the network connection for Astra Linux using the command ping <ip address>

```
[root@Astra ~]# ping 192.168.1.68
PING 192.168.1.68 (192.168.1.68) 56(84) bytes of data.
64 bytes from 192.168.1.68: icmp_seq=1 ttl=64 time=0.118 ms

64 bytes from 192.168.1.68: icmp_seq=2 ttl=64 time=0.358 ms
64 bytes from 192.168.1.68: icmp_seq=3 ttl=64 time=0.055 ms
```

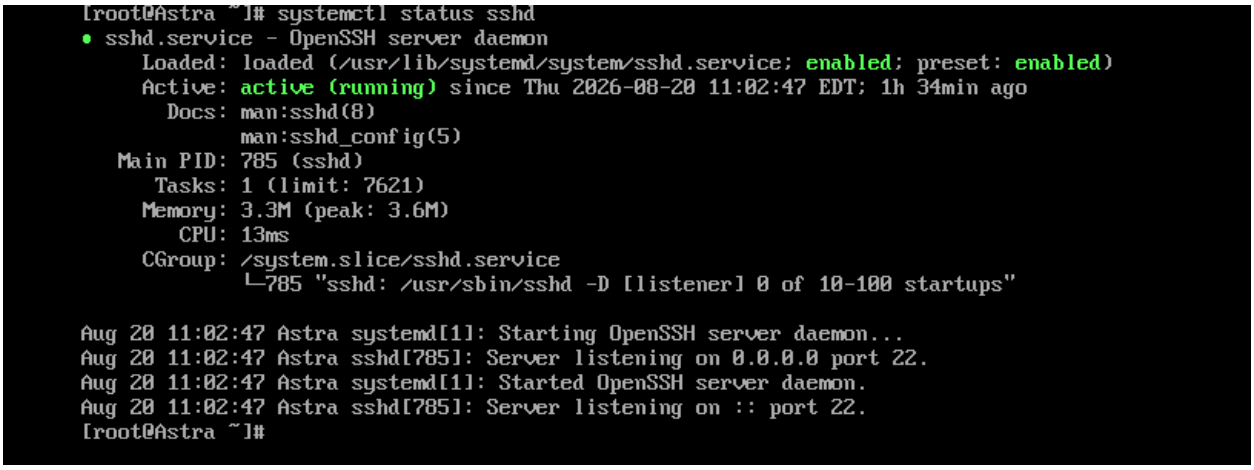
To manully connect to network in astra linux type ‘nmtui’ command





Step 3: Configure SSH

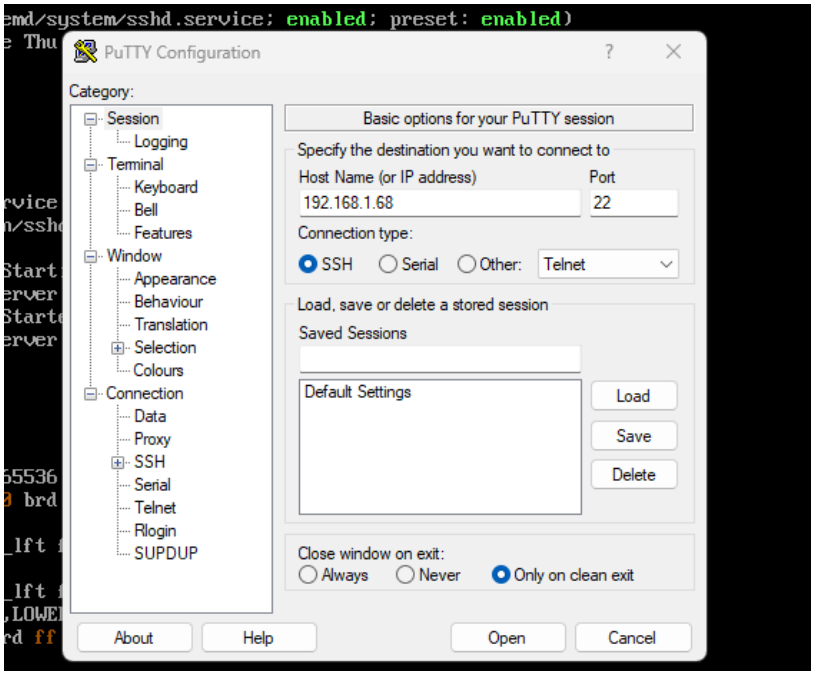
-To install the ssh serivce package “ yum install –y openssh-cilent

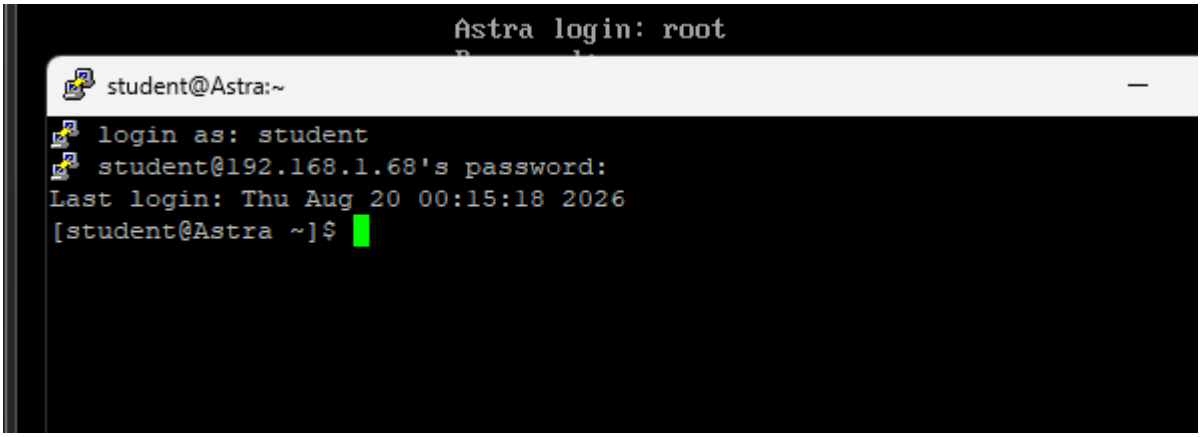


-Connected Two systems using ssh

Step 4 :Connect Using Putty

-To Connect to putty , provide the ip address of the system.





PHASE 2

Task 1- Create and Configure Your User Account

- **Create a new user account using your own name**

  - To a create a new user we use the command : sudo userdd <username>
  - To Verify whether the user is created or not, just type: id <username> , You can see the details like username and group it belongs to

```
[root@Astra ~]#
[root@Astra ~]# useradd hrishi
[root@Astra ~]# id hrishi
uid=1001(hrishi) gid=1001(hrishi) groups=1001(hrishi)
[root@Astra ~]# _
```

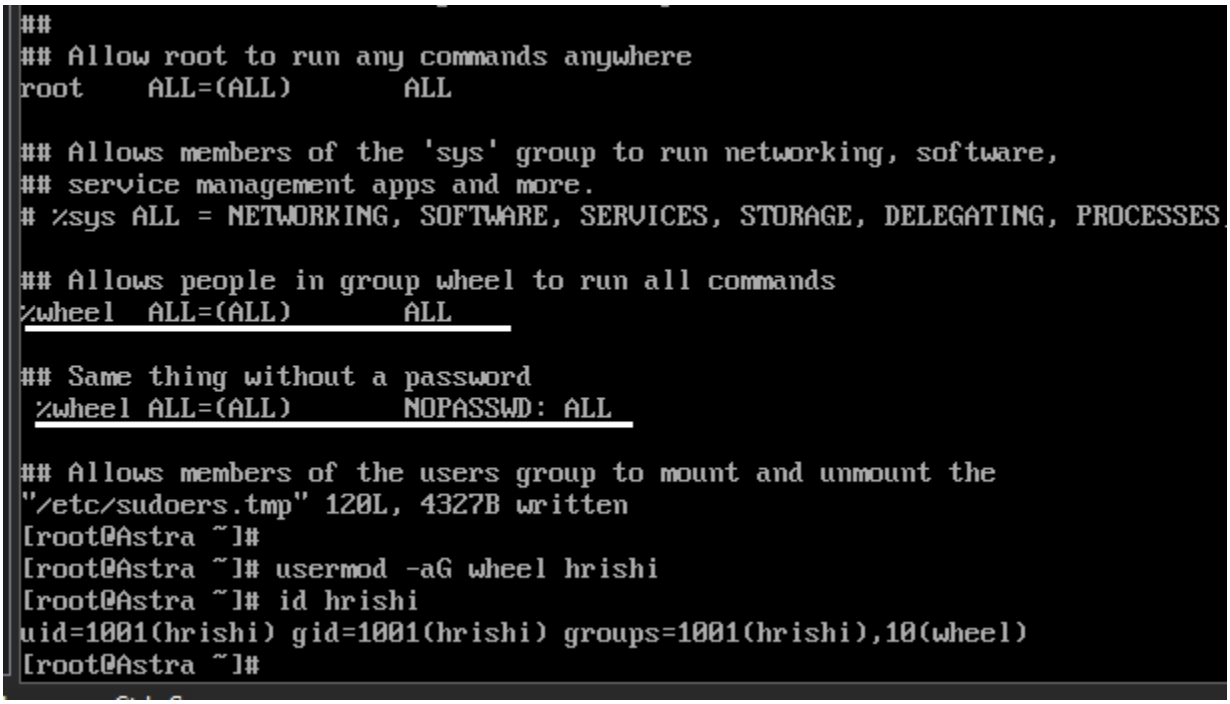
- **The user account must have sudo privileges without requiring a password**

  - For a user to have sudo privileges the user either should be in the sudoers file or The user should be a member of wheel group
  - To add a user to wheel group type : usermod –aG username wheel
  - Then you go to the configuration file named sudoers and edit it (Enable the wheel group no passwd(all) comment), or just manually enter the instruction.

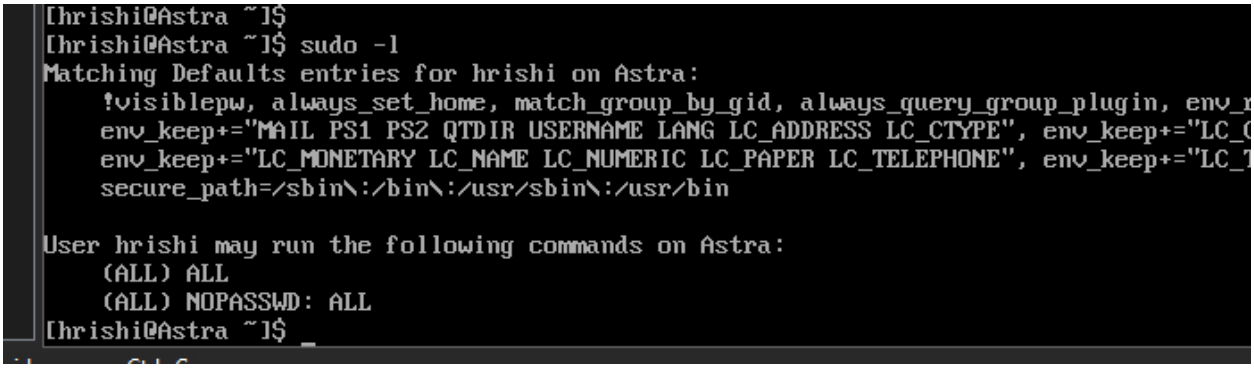
```
[root@Astra ~]#
[root@Astra ~]# vim /etc/sudoers
-bash: vim: command not found
[root@Astra ~]# visudo
visudo: /etc/sudoers busy, try again later
[root@Astra ~]#
[root@Astra ~]# jobs
[1]+  Stopped                 visudo
[root@Astra ~]# visudo
visudo: /etc/sudoers busy, try again later
[root@Astra ~]# jobs_
```

Vim package was not installed , so i used visudo .

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- Configured the sudoers file and added the user to the wheel group. And verifed the whether the added user has sudo privilege or not.



- **This user account has been created specifically for this practical event. Therefore, configure the account to automatically expire on December 31, 2026.**

  -To make changes of the user we use the command “chage” , which means change. -To change the expiry date the command is – chage –E <date> <username>
  -We can verify it through the command – chage –l username

```
[root@Astra ~]# chage -E 2026/12/31 hrishi
[root@Astra ~]# chage -l
Usage: chage [options] LOGIN

Options:
  -d, --lastday LAST_DAY        set date of last password change to LAST_DAY
  -E, --expiredate EXPIRE_DATE  set account expiration date to EXPIRE_DATE
  -h, --help                    display this help message and exit
  -i, --iso8601                 use YYYY-MM-DD when printing dates
  -I, --inactive INACTIVE       set password inactive after expiration
                                to INACTIVE
  -l, --list                    show account aging information
  -m, --mindays MIN_DAYS        set minimum number of days before password
                                change to MIN_DAYS
  -M, --maxdays MAX_DAYS        set maximum number of days before password
                                change to MAX_DAYS
  -R, --root CHROOT_DIR         directory to chroot into
  -W, --warndays WARN_DAYS      set expiration warning days to WARN_DAYS

[root@Astra ~]# chage -l hrishi
Last password change                                    : Aug 19, 2026
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : Dec 31, 2026
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
[root@Astra ~]#
```

Task 2 – Create and Configure the /astra Directory

- **Create a directory named: Astra**
  - mkdir /Astra

- **The Directory must be configured so that any user can create files or directories inside it,but only the user who owns a particular file or directory can delete it.**
-
  - For the user to have the only privilege to delete a file , the File/Directory must have a special permission called sticky bit. And its value is 1.

- **Configuring the Appropriate Permission**

```
Last login: Wed Aug 19 06:46:50 on tty1
[root@Astra ~]# mkdir /Astra
[root@Astra ~]# chmod 1777 /Astra
```

- **Verifying and Permission Demonstration**

```
[root@Astra ~]#
[root@Astra ~]# touch /Astra/newFile.txt
[root@Astra ~]# sudo -su student
[student@Astra root]$ cd
[student@Astra ~]$ rm -rf /Astra/newF[  411.274150] atkbd serio0: Spurious
ile.txt            cd
[student@Astra ~]$ rm -rf /Astra/newFile.txt
rm: cannot remove '/Astra/newFile.txt': Operation not permitted
[student@Astra ~]$ _
```

-Created a file inside the directory, then switched to another user to check if the permission works and verified that other users cannot delete the file.

Task –3 Monitoring and Activity log

- **Creating File named monitoring and the file should have store the log informations.**
  -First to create a custom log we need to first get into rsyslog.d file in the /etc , directory
  - Inside the '/rsyslog.d' we need to create a new file with extension .conf eg:newFile.conf . What I did here was i created a configuration file to store info logs.
  -Then we open this creatred file and add the log details and file name, location.
  - The rsyslog log services should be restarted after editing the configuration file.

```
yum install -y vim
clear
vim /etc/rsyslog.d/monitoring.conf
systemctl restart rsyslog
```

- The text in the below figure is added inside the 'monitoring.conf' - configuration file , what it does is that it stores the information logs to monitoring.txt fiile inside the log directory

```
*.info /var/log/monitoring.txt
~
~
~
```

- Configuring the logging mechanism.
  -We use the command logger to generate a test log entry .The synopsis is Logger –p authpriv.info "msg". I check it with the 'man' command.
  -This will create alog file, we can view the logs with the "cat" command.

```
[root@Astra ~]#
[root@Astra ~]# logger -p authpriv.info "this is a sample log 2"
[root@Astra ~]# cat /var/log/monitoring.txt
Aug 20 00:52:51 Astra systemd[1]: Stopping System Logging Service...
Aug 20 00:52:51 Astra rsyslogd[809]: [origin software="rsyslogd" swVersion="8.2506.0-2.el9" x-p
Aug 20 00:52:51 Astra systemd[1]: rsyslog.service: Deactivated successfully.
Aug 20 00:52:51 Astra systemd[1]: Stopped System Logging Service.
Aug 20 00:52:51 Astra systemd[1]: Starting System Logging Service...
Aug 20 00:52:55 Astra rsyslogd[10400]: [origin software="rsyslogd" swVersion="8.2506.0-2.el9" x
Aug 20 00:52:55 Astra rsyslogd[10400]: imjournal: journal files changed, reloading...  [v8.2506
Aug 20 00:52:55 Astra systemd[1]: Started System Logging Service.
Aug 20 00:54:08 Astra root[10405]: this is a sample log
Aug 20 00:57:32 Astra root[10419]: this is a sample log 2
[root@Astra ~]# _
```

- This will show the recently made logs . The required log information is shown in the above figure.

Task 4 – Configure a Graphical Environment

There are no gui currently configured in the astra linux , So after researching I found that the typical desktop environmrnt for astra linux is FLY.

- **Installing the Required Packagest**

  -Firstly check the os version.

```
[root@Astra ~]# cat /etc/os-release
NAME="Astra Linux"
VERSION="9.7 (Blue Onyx)"
ID="rocky"
ID_LIKE="rhel centos fedora"
VERSION_ID="9.7"
PLATFORM_ID="platform:el9"
PRETTY_NAME="Astra Linux 9.7 (Blue Onyx)"
ANSI_COLOR="0;32"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:rocky:rocky:9::baseos"
HOME_URL="https://rockylinux.org/"
VENDOR_NAME="RESF"
VENDOR_URL="https://resf.org/"
BUG_REPORT_URL="https://bugs.rockylinux.org
SUPPORT_END="2032-05-31"
ROCKY_SUPPORT_PRODUCT="Rocky-Linux-9"
ROCKY_SUPPORT_PRODUCT_VERSION="9.7"
REDHAT_SUPPORT_PRODUCT="Rocky Linux"
REDHAT_SUPPORT_PRODUCT_VERSION="9.7"
[root@Astra ~]# _
```

  -

  -Check avaliable desktop environment groups

  -Then install the graphical environment

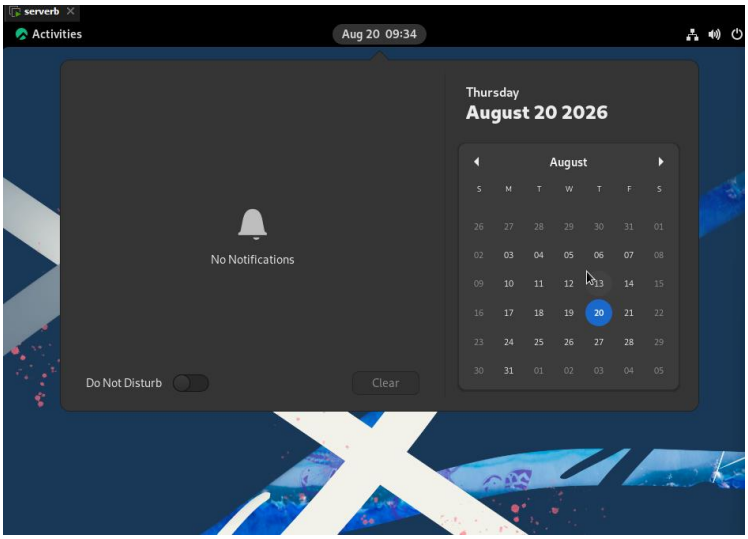
```
  68  yum group list installed
  69  yum group list
  70  dnf groupinstall "Server with GUI" -y
  71  history'
  72  history
root@Astra ~]#
```

* **Starting GUI**

-configure rocky linux to boot into the GUI
-Reboot the system to launch the GUI

```
  72  history
[root@Astra ~]# systemctl get-default
multi-user.target
[root@Astra ~]# sudo systemctl set-default graphical.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
[ 3944.598978] systemd-rc-local-generator[67256]: /etc/rc.d/rc.local is not marked executable, skipping.
[root@Astra ~]# systemctl get-default
graphical.target
[root@Astra ~]# _
```

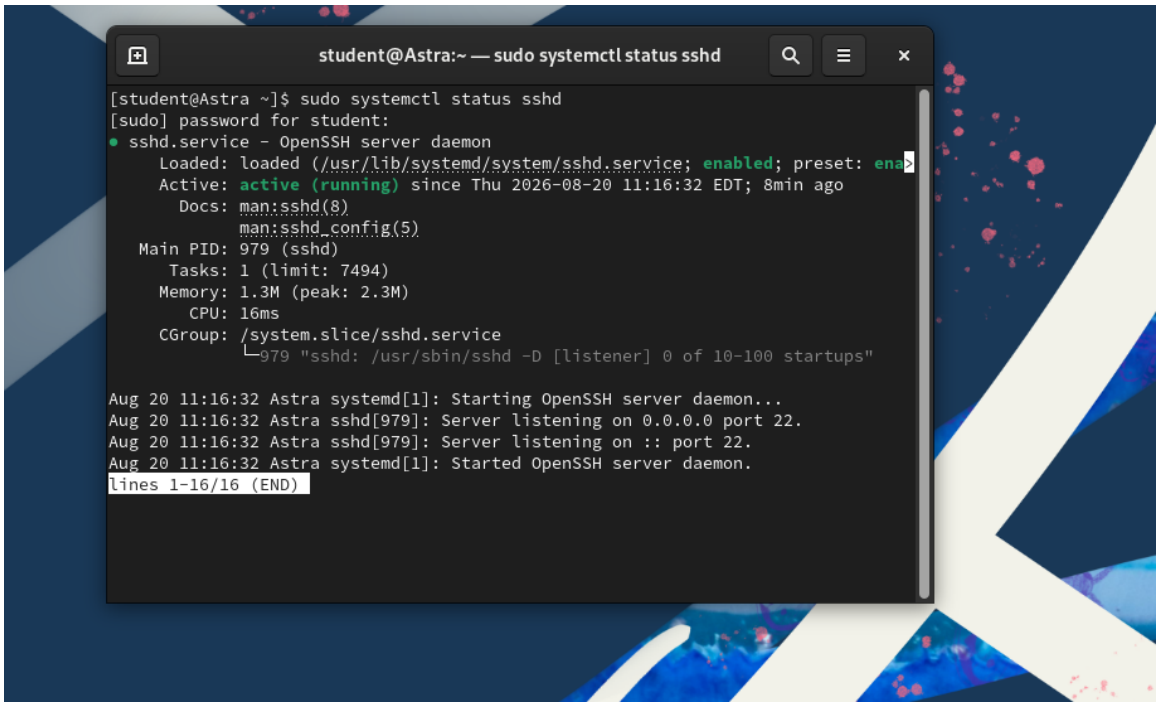
-System can be used through the GUI & required graphical services are running smoothly.



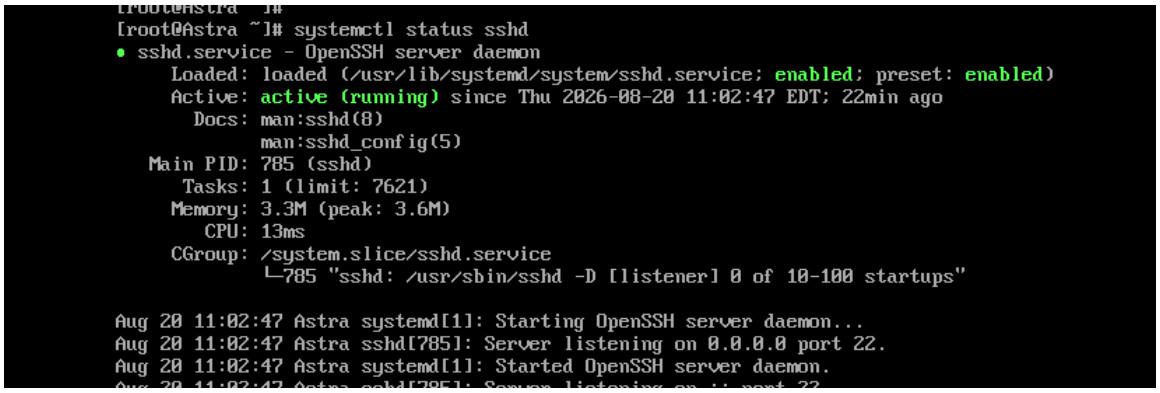
Task 5 –Remote Desktop accesss

- **Configure the required desktop compenonts.**
  - **To** establish connection between a local and a remote system both system should have ssh service.

  - System 1

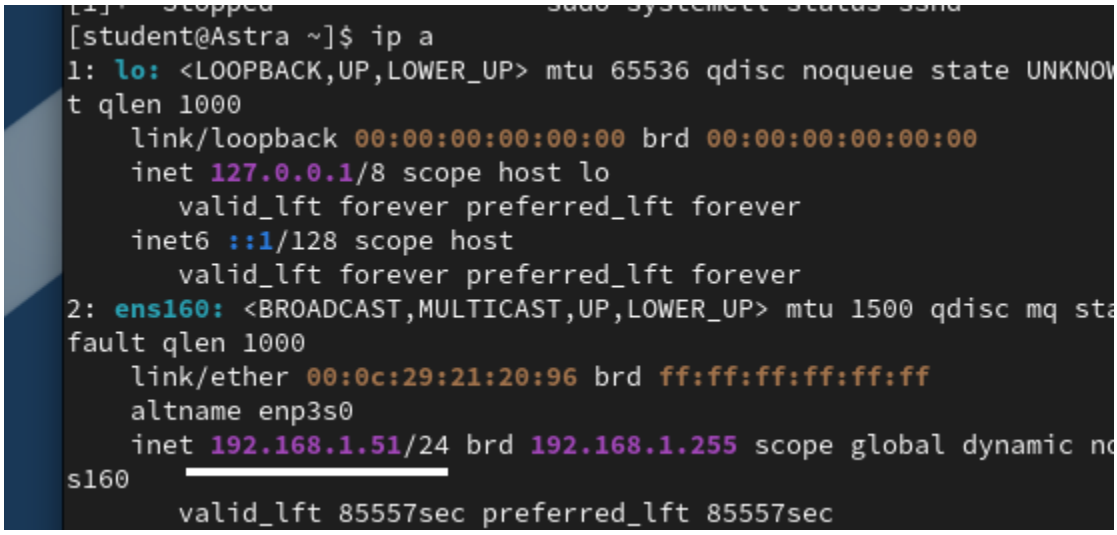


System 2:

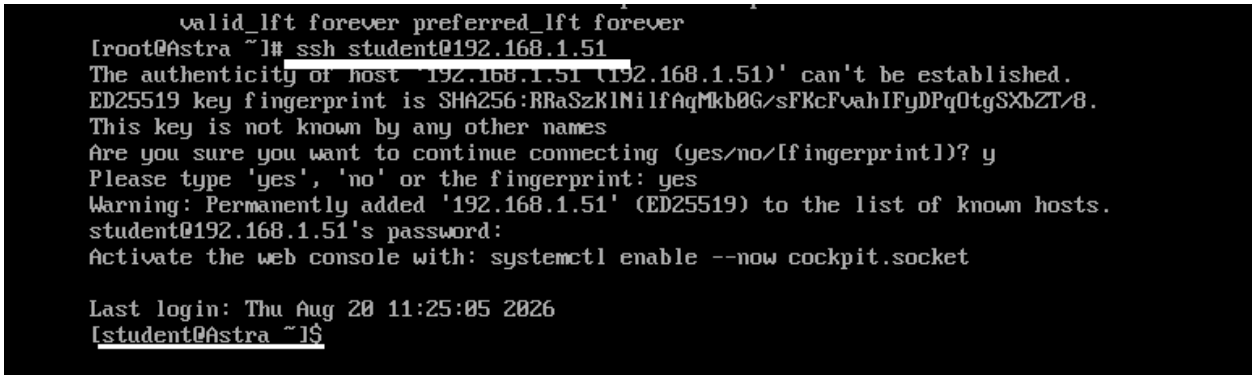


-Since sshd is available in both system thats pretty much it , no need of additional services.

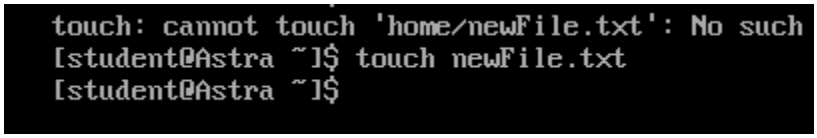
-To establish a secure connection both host and remote server should be in a same network. To see it check the ip (ping <ip address>)



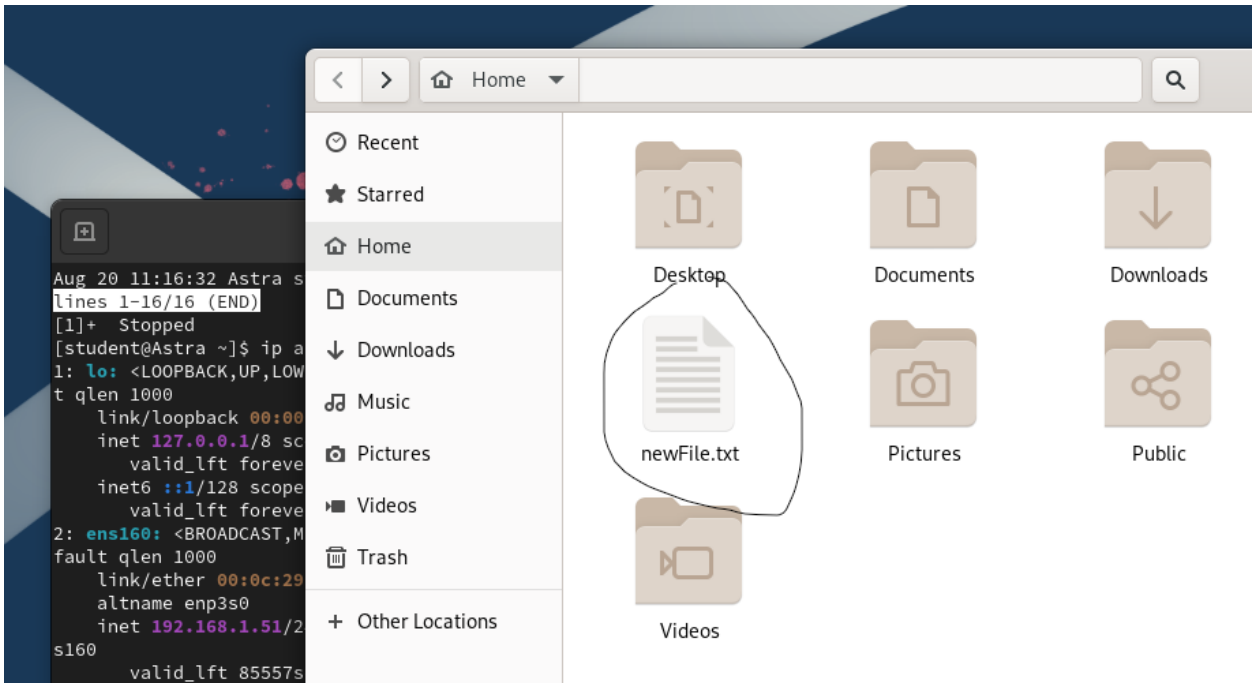
-To remotely access a system we use the ssh command



Verifying whether the Astra graphical desktop can be accessed .



Created a file from local machine.



-The file is formed in the remote system.